

CURRICULUM VITAE

CONTACT INFORMATION

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EDUCATION

- **Ph.D., Mechanical Engineering (Control Systems)**
Southern Illinois University Carbondale (SIUC), 2026
Dissertation: *Lyapunov–Krasovskii Based Approach to Time-Delay Systems: Computational Efficiency and Feedback Control*
- **M.S., Mechanical Engineering (Control Systems)**
Southern Illinois University Edwardsville (SIUE), 2020
- **B.E., Mechanical Engineering**
Ho Chi Minh City University of Technology and Education (HCMC UTE), 2013

ACADEMIC AND PROFESSIONAL EXPERIENCE

Randolph College, Lynchburg, VA

Assistant Professor, Robotics and Mechatronics Engineering Jun. 2026 – Present

- Teach and develop undergraduate courses in robotics, mechatronics, mechanical engineering, control systems, sensors, actuators, and related engineering areas.
- Contribute to curriculum development, laboratory development, student mentoring, and program assessment activities supporting engineering education.
- Support ABET-related instructional and program outcomes through project-based learn-

ing, laboratory activities, and continuous course improvement.

Southern Illinois University Edwardsville (SIUE), Edwardsville, IL

Instructor of Record / Graduate Teaching Assistant, Part-time

2018 – 2026

- Independently taught or supported undergraduate and graduate engineering courses, including Numerical Simulation, Introduction to Mechatronics, Sensors and Actuators, and Optimal Control.
- Developed syllabi, lectures, assignments, laboratory exercises, examinations, MATLAB projects, hardware demonstrations, and grading materials.
- Mentored students in theoretical, computational, and experimental engineering activities involving control systems, mechatronics, robotics, and numerical simulation.

Southern Illinois University Edwardsville / Southern Illinois University Carbondale

Graduate Research Assistant, Full-time

2020 – 2026

- Conducted research on Lyapunov–Krasovskii functionals, stability analysis, feedback control, invariant subspace methods, and computational approaches for time-delay systems.
- Developed decomposition-based frameworks for efficient stability and control analysis of systems with delays and uncertainties.
- Published research in leading control journals, including *IEEE Transactions on Automatic Control* and *International Journal of Robust and Nonlinear Control*.

TuSimple – Autonomous Trucking, Tucson, AZ

Sensor Hardware Engineer Intern, Full-time

May 2022 – Aug. 2022

- Evaluated LiDAR, camera, radar, and IMU systems for Level-4 autonomous trucks under diverse operating conditions.
- Characterized LiDAR range accuracy, field of view, mirror noise, blooming noise, IMU Allan variance, and camera performance.
- Conducted sensor benchmarking using MATLAB, Python, CloudCompare, ROS, CAN, serial communication, and AWS-based data processing.

Kawahara Co., Ltd., Nagoya, Japan

Mechanical Engineer, Full-time

Oct. 2016 – Jun. 2017

- Managed quality control for precision automotive components and maintained CNC systems to ensure production accuracy.

Sakae Seiko Co., Ltd., Obu, Japan

Mechanical Engineer, Full-time

Jun. 2015 – Sep. 2016

- Designed and programmed precision machining paths using Mastercam and SolidWorks for mold manufacturing.

PROFESSIONAL CREDENTIALS, CERTIFICATIONS, OR LICENSING

- Professional engagement through IEEE, control systems, robotics, and engineering education publication and presentation venues.
- Technical experience in autonomous systems, sensor evaluation, LiDAR, IMU, camera systems, CAN communication, ROS, MATLAB, Python, mechanical design, and mechatronics.

PROFESSIONAL DEVELOPMENT ACTIVITIES

- Developed instructional materials and laboratory activities for undergraduate and graduate engineering courses in numerical simulation, mechatronics, sensors and actuators, and optimal control.
- Developed hands-on laboratory modules using MATLAB, Simulink, Arduino, LabVIEW, ROS, NVIDIA Jetson, LiDAR, IMU, encoders, cameras, and real-time data acquisition.
- Completed industry-based professional practice in autonomous vehicle sensor evaluation, including LiDAR, camera, radar, IMU, CAN communication, ROS-based acquisition, and cloud-based data processing.
- Participated in scholarly dissemination through peer-reviewed journal publications and conference presentations in control systems, time-delay systems, robotics, autonomous systems, and engineering education.

SELECTED PUBLICATIONS AND PRESENTATIONS

- H. Phan-Van and K. Gu (2024), “Structured Invariant Subspace and Decomposition of Systems with Time Delays and Uncertainties,” *International Journal of Robust and Non-linear Control*, 34(10), 6480–6512.
- K. Gu and H. Phan-Van (2023), “External Direct Sum Invariant Subspace and Decomposition of Coupled Differential–Difference Equations,” *IEEE Transactions on Automatic Control*, 69(2), 1022–1028.

- H. Phan-Van, H. Ferdowsi, and N. Lotfi (2020), “Trajectory Control of Omnidirectional Mobile Robots Considering Low-Level Actuator Dynamics,” *28th Mediterranean Conference on Control and Automation (MED)*, 996–1001.
- N. Lotfi, J.A. Novosad, and H. Phan-Van (2019), “A Multidisciplinary Course and Laboratory Platform for Teaching the Fundamentals of Advanced Autonomous Vehicles,” *ASEE Annual Conference & Exposition*.

RESEARCH EXPERIENCE

Time-Delay Systems and Feedback Control

SIUE/SIUC, 2020 – 2026

- Developed Lyapunov–Krasovskii-based methods for stability analysis and feedback control of time-delay systems.
- Proposed invariant-subspace and external-direct-sum decomposition methods to reduce computational complexity in delayed systems.
- Extended theoretical methods to uncertain and multi-delay systems and validated results through numerical simulation.

Autonomous Mobile and Agricultural Robots

SIUE, 2018 – 2026

- Designed and built omnidirectional mobile robots and agricultural robotic platforms integrating LiDAR, IMU, camera systems, embedded controllers, and real-time software.
- Implemented trajectory tracking and Lyapunov-based control strategies for robotic systems under uncertainty.
- Applied robotic platforms to laboratory-based teaching in mechatronics, sensors and actuators, autonomous systems, and control.

Educational Robotics and Course Development

SIUE, 2018 – 2026

- Co-developed a multidisciplinary course and laboratory platform for teaching the fundamentals of autonomous ground vehicles.
- Built teaching robots using Arduino, NVIDIA Jetson, sensors, actuators, and ROS-based tools for SLAM, localization, and control laboratories.
- Published related engineering education work in the *ASEE Annual Conference & Exposition*.

Sensor Evaluation and Data-Driven Testing

TuSimple, 2022

- Conducted LiDAR, IMU, radar, and camera testing for autonomous truck perception systems.
- Applied Allan variance, NEDT, LiDAR range testing, field-of-view analysis, and data-driven benchmarking for sensor reliability evaluation.

Robotic Fish

HCMC UTE, 2011 – 2013

- Designed a biomimetic robotic fish and studied actuator synchronization and hydrodynamic performance.

TEACHING EXPERIENCE

As an **independent instructor** and teaching assistant at Southern Illinois University Edwardsville (SIUE), I taught and supported undergraduate and graduate engineering courses. My instructional experience includes course preparation, lecture delivery, laboratory development, assessment, grading, and student mentoring in theoretical, computational, and experimental engineering topics.

- **ME 354 – Numerical Simulation:** Sole instructor for MATLAB-based numerical analysis; developed lectures, projects, and exams covering curve fitting, numerical integration, and ordinary differential equation solving.
- **ME 358 – Introduction to Mechatronics:** Developed hands-on laboratory modules integrating Arduino, LabVIEW, sensors, actuators, and real-time data acquisition.
- **ME 320 – Sensors and Actuators:** Supported laboratory-based instruction using LiDAR, IMU, encoders, and sensor modeling to connect measurement systems with control applications.
- **ME 563 – Optimal Control:** Developed course materials involving LQR, Lyapunov-based design, MATLAB simulation, and trajectory control of robotic systems.

I was recognized with the *Outstanding Teaching Assistant Award* at SIUE in 2020.

SERVICE ACTIVITIES

- Mentored undergraduate and graduate students in robotics, mechatronics, control systems, sensors, and numerical simulation projects.
- Supported laboratory development for courses involving sensors, actuators, robotics, autonomous systems, and control.

- Contributed to engineering education through development of robotics-based instructional platforms and publication in ASEE.
- Supported student learning through project-based instruction, hardware demonstrations, MATLAB simulations, and laboratory supervision.

HONORS & AWARDS

- Outstanding Teaching Assistant Award, SIUE (2020)
- Research Grants for Graduate Students (RGGS), SIUE

TECHNICAL SKILLS

- **Control Systems:** Lyapunov stability, Lyapunov–Krasovskii functionals, time-delay systems, feedback control, optimal control, LMI-based methods
- **Robotics and Mechatronics:** Omnidirectional robots, agricultural robots, autonomous ground vehicles, sensors and actuators, embedded systems
- **Sensors:** LiDAR, camera, radar, IMU, encoders, sensor benchmarking, sensor calibration, CAN communication
- **Programming:** MATLAB, Python, C/C++
- **Tools & Platforms:** Simulink, ROS, CAN, AWS, Raspberry Pi, NVIDIA Jetson, LabVIEW, Arduino
- **Mechanical Design:** SolidWorks, Mastercam, AutoCAD